



TBML Enters the “Stablecoin Settlement Layer” Era: 2025 Observation Brief

Federal Financial Education Institute (FFEdU)

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Purpose & Scope. This brief provides a risk-analytic framework for understanding how trade-based money laundering (TBML) is evolving as stablecoins become a de facto settlement instrument in certain cross-border value-transfer ecosystems. It synthesizes (i) official U.S. typologies and implementation signals, (ii) measured on-chain risk trends, and (iii) emerging perimeter debates around settlement-layer identifiability, to equip readers with cross-ledger coherence tests spanning trade documentation, logistics corroboration, banking flows, and on-chain settlement traces. The brief is designed for education and policy literacy; it is not legal advice and does not provide operational instructions.

Thesis. TBML is increasingly coupling with stablecoin-mediated settlement, accelerating value-transfer velocity and composability while pushing U.S. policy attention upward from “on/off-ramps” toward settlement-layer identifiability, accountability, and supervisory visibility.

I. Executive Summary

1) TBML is “value transfer explained as trade”

TBML is best conceptualized as value transfer camouflaged as legitimate commerce. The trade leg—documentation, pricing, routing, counterparties, and payment structure—serves as the explanatory wrapper that can obscure the economic substance: source, ownership/control, and destination of value.



2) What changed in 2025 (and what the record supports)

- Official typologies tightened. FinCEN’s 2025 advisory emphasizes repeatable risk structures in which Mexico-based transnational criminal organizations (TCOs) leverage professional laundering networks (including those characterized as “Chinese money laundering networks”) to launder illicit proceeds, and it supplies red flags and BSA reporting expectations. These materials support typologies and indicators, not a single universal “closed-loop” pipeline.^{1 2}
- Stablecoins are now a first-order settlement variable. Major analytics firms report that stablecoins dominate measured illicit on-chain value transfer and that certain low-fee ecosystems can support higher transaction velocity and fragmentation (layering), compressing investigative timelines and increasing the importance of cross-domain attribution and coherence testing.^{3 4}
- The stablecoin perimeter is being operationalized. The GENIUS Act (enacted July 18, 2025) establishes a federal framework for payment stablecoins, while Treasury’s 2025 implementation notice highlights financial-integrity and sanctions considerations as core design questions.^{5 6}
- Settlement-layer anonymity is emerging as a policy target. In December 2025, the Stablecoin Standards Authority (SSA) publicly announced a petition urging coordinated GENIUS implementation, heightened risk treatment for USDT, and a concept of “on-chain node KYC” / a node registry—an agenda signal that the perimeter debate may move from issuers and intermediaries toward identifiable infrastructure.^{7 8}

3) FFEdU’s two-line toolkit (what to “take away”)

- Four Ledgers: Trade / Logistics / Banking / On-chain
- Four Mismatches: Tempo / Path / Identity / Geography

Educational claim. In the stablecoin settlement-layer era, TBML risk is most reliably surfaced when these four ledgers cannot be reconciled into a single commercially plausible story.

II. Evidence Grading & Methodological Guardrails

A. Evidence grading (what each source can—and cannot—support)



FFEdU uses an implicit evidence hierarchy to prevent over-claiming:

- **E1 (Highest):** Statutes, Federal Register notices, formal agency advisories/guidance.

Supports: legal boundaries, official definitions, regulatory intent, supervisory expectations.

- **E2:** Indictments, judgments, sanctions designations, official enforcement releases.

Supports: case-level chains and proven methods in those cases.

- **E3:** Quantitative analytics research (e.g., crypto crime reports).

Supports: measured trends in attributed/identified activity, subject to visibility and methodology limits.

- **E4:** Media investigations, interviews, advocacy petitions/white papers.

Supports: hypotheses, agenda signals, leads; not a substitute for E1/E2 factual grounding.

B. Methodological guardrails (discipline rules)

1. **Typology $\neq$ single pipeline.** Official advisories and risk indicators describe repeatable structures and reporting triggers, not a universal mechanism that always operates end-to-end.^{1 2}
2. **Correlation $\neq$ causation.** Rising stablecoin shares in measured illicit on-chain activity do not imply stablecoins “cause” TBML; the narrower claim is that stablecoins can reduce settlement friction and increase velocity/fragmentation.^{3 4}
3. **On-chain transfer $\neq$ legal/economic finality.** Redemption constraints, freezing authority, sanctions exposure, and custodial banking risk affect the ability to realize and deploy value.
4. **Red flags $\neq$ adjudication.** Indicators trigger enhanced diligence and reporting consideration; they are not proof of wrongdoing.

III. Coherence Framework (FFEdU): “Four Ledgers, One Story”

Quoted definition (FFEdU Coherence Standard).



In the stablecoin settlement-layer era, TBML risk is best detected not by any single dataset, but by cross-ledger incoherence—when trade documentation, logistics corroboration, banking/payment flows, and on-chain settlement traces cannot be reconciled to a commercially plausible narrative with attributable control.

The four ledgers.

1. **Trade Ledger (Documents & Pricing).** Contracts, invoices, Incoterms, product description, unit pricing, quantity, counterpart roles.
2. **Logistics Ledger (Movement & Corroboration).** Bills of lading/airway bills, carrier data, container events, warehouse/receiving evidence, customs declarations, route plausibility.
3. **Banking Ledger (Payments & Counterparties).** Payment path, third-party payments, beneficial ownership, account behavior, currency conversions, timing vs. milestones.
4. **On-Chain Ledger (Settlement & Address Attribution).** Stablecoin transfers, aggregation/fragmentation patterns, counterparty clusters, exchange exposure, cadence vs. business cycle, attribution confidence.

What “coherence” demands (high-level).

- **Commercial plausibility:** price, routing, counterpart roles, and margins make business sense.
- **Temporal plausibility:** payments and settlement timing align with shipment/invoicing/acceptance/terms.
- **Control plausibility:** the legal persons controlling accounts and addresses can be identified and documented.
- **Narrative consistency:** the same story is supported by trade docs, logistics facts, banking flows, and on-chain traces.

IV. 2025 Pivot Mechanism: Why This Year Matters (Beyond a Timeline)

2025 marks a **mechanistic pivot** created by the interaction of (1) legal perimeter construction, (2) typology formalization, and (3) settlement-layer agenda formation.



A. Perimeter construction (GENIUS): stablecoins become governable payment infrastructure

The GENIUS Act’s enactment and Treasury’s implementation process move payment stablecoins toward a more explicit federal perimeter.^{5 6} As stablecoins are treated as payment-adjacent infrastructure, integrity controls (BSA/AML and sanctions) become design variables, not optional overlays. The practical effect is an institutional willingness to ask: *where must supervisory visibility attach to preserve settlement integrity?*

B. Typology formalization (FinCEN): professional laundering becomes compliance vocabulary

FinCEN’s 2025 advisory strengthens typology-based detection by describing common features, indicators, and reporting expectations.^{1 2} The key educational implication is methodological: typologies help institutions identify **coherence failures** early—without assuming any single “pipeline” is universal.

C. Agenda formation (SSA): the debate shifts upward to settlement-layer identifiability

SSA’s December 2025 petition is not a rule; it is an agenda signal.^{7 8} But it is consequential: it frames “settlement-layer anonymity” as an implementation gap and proposes “on-chain node KYC” / a node registry concept. If such framing gains traction, compliance emphasis may climb the stack—from issuers and intermediaries toward identifiable settlement infrastructure—with second-order effects on cost, corridor availability, and risk migration.

Bottom line (the pivot). TBML risk management is increasingly about who can credibly tell a coherent cross-ledger story—and whether the settlement layer itself becomes a regulated identity surface.

V. Terminology Boundaries (TBML / IVTS / BMPE)

- **TBML (Trade-Based Money Laundering):** Value transfer using trade activity—real or fabricated—via mispricing, document manipulation, routing complexity, third-party payments, or supply-chain layering to disguise origin/ownership/destination.^{9 10}



- **IVTS (Informal Value Transfer Systems):** Informal value transfer (e.g., underground remittance, “matching,” off-ledger settlement) that does not require trade. IVTS may intersect with TBML but is not synonymous with TBML.
- **BMPE (Black Market Peso Exchange):** A classic demand-matching typology that can overlap with TBML and IVTS but should be used precisely.

Practical rule.

No meaningful trade-document/pricing/logistics element → **IVTS** is usually the appropriate frame.

Trade documentation/pricing/logistics is the disguise mechanism → **TBML**.

Demand matching with trade-based value return is central → **BMPE**.

VI. What the Public Record Supports (and What It Does Not)

A. What can be stated rigorously from E1 sources

FinCEN’s advisory supports that Mexico-based TCOs have used professional laundering networks—described as CMLNs—to launder proceeds, and it provides red flags and BSA reporting expectations.^{1 2}

What should not be asserted from E1 alone: a deterministic chain specifying particular commodities, jurisdictions, balancing mechanisms, or settlement assets as universal. Those claims are case-dependent and typically require E2 materials to state as fact.

B. Stablecoin settlement: the measured claim and its limit

Measured analytics indicate stablecoins constitute a dominant share of identified illicit on-chain transaction volume, and that certain low-fee ecosystems appear prominently in attributed illicit flows.^{3 4}

Limit: “identified illicit volume” is not “all illicit volume”; attribution and observability constraints apply.



VII. Red-Flag Bundles: Four “Mismatch Packs”

How to use these bundles. A single indicator is rarely dispositive. Risk escalates when multiple indicators appear together and cannot be reconciled to a plausible commercial narrative.

Pack A — Tempo Mismatch (Settlement vs. Business Cycle)

Signals (cluster):

- On-chain transfers occur at a cadence inconsistent with invoicing, shipment milestones, acceptance, or payment terms.
- Repeated short-cycle fragmentation and reconsolidation without a business explanation.
- Settlement spikes immediately before/after trade milestones that are otherwise weakly evidenced.

Why it matters: TBML often surfaces as a timing story optimized for movement rather than commerce.

Education-level next step: Reconstruct a single timeline across the four ledgers (ship → invoice → acceptance → pay) and test whether settlement behavior plausibly follows the contract terms.

Pack B — Path Mismatch (Who Pays Whom, and Why)

Signals (cluster):

- Third-party payments not aligned with trade roles, lacking credible written commercial rationale.
- Unnecessary complexity in payment routing (multiple hops/currencies/institutions) without supply-chain necessity.
- Payment counterparties that do not match contractual counterparties or control persons.

Why it matters: Path complexity is a classic layering vector—especially when it cannot be justified by trade mechanics.



Education-level next step: Require a documented “payment logic map” (payer → payee → reason) and verify that beneficial ownership/control is consistent across trade counterparties and payment participants.

Pack C — Identity/Control Mismatch (Attribution Failure)

Signals (cluster):

- Beneficial ownership/control information is missing, unstable, or inconsistent across trade entities, bank accounts, and on-chain addresses.
- Address attribution confidence is low, shifting, or inconsistent with declared roles.
- Repeated use of intermediaries where control cannot be documented.

Why it matters: In stablecoin-enabled settlement, the critical risk is often not traceability per se but attribution and control across domains.

Education-level next step: Establish a minimum attribution standard: identify controlling persons; document authority to transact; retain records linking legal persons to relevant accounts and addresses.

Pack D — Geography & Sanctions-Exposure Mismatch (Corridor Plausibility)

Signals (cluster):

- Trade routes and settlement corridors combine jurisdictions in ways lacking commercial plausibility.
- Recurrent exposure to elevated-risk jurisdictions, sanctions touchpoints, or unexplained “corridor switching.”
- Logistics corroboration is thin or inconsistent with claimed routes.

Why it matters: Jurisdictional seams are frequently exploited to create deniability and degrade visibility.



Education-level next step: Test corridor plausibility against trade basics (routing, shipping time, carrier evidence) and overlay sanctions/geographic risk to determine whether escalation is warranted.

VIII. A “Public-Materials” Micro-Example (Structured as a 1-Minute Classroom Exercise)

Exercise prompt (structure only).

A trading entity presents facially consistent trade documents. Logistics corroboration is partial but not conclusively false. Yet the **payment and settlement story cannot be reconciled**: funds originate from an unrelated third party; the banking path includes unnecessary currency hops; and on-chain settlement shows repeated fragmentation and rapid reconsolidation on a cadence inconsistent with shipment and acceptance milestones. Beneficial ownership documentation does not credibly link controlling persons to both bank accounts and on-chain addresses.

Question: Which of the Four Mismatch Packs are implicated, and which of the Four Ledgers fail coherence?

Education-level answer (no operational detail): This presents a cross-ledger coherence failure (Tempo + Path + Identity), justifying enhanced diligence, corroboration requests, and—where legally appropriate—reporting consideration consistent with typology-based guidance.^{1 2}

Rule: If you cannot reconcile the Four Ledgers into One Story, you have a coherence failure—escalate scrutiny.

IX. Policy Watch: From Entity Compliance Toward Infrastructure Visibility

A. GENIUS implementation and integrity-by-design



GENIUS and Treasury’s implementation process situate payment stablecoins within a framework where prudential soundness and financial integrity (including sanctions/AML) are design constraints.^{5 6}

B. Settlement-layer anonymity as a policy object

SSA’s December 2025 petition frames “settlement-layer anonymity” as an implementation gap and advances “on-chain node KYC” / a node registry concept.^{7 8}

Educational implication: if infrastructure identifiability becomes a regulatory lever, stablecoin settlement may exhibit greater compliance centralization, changing corridor economics and potentially displacing certain risk into less visible channels.

Footnotes (Verifiable Links)

1. FinCEN Advisory FIN-2025-A003 (Aug. 28, 2025), “FinCEN Advisory on the Use of Chinese Money Laundering Networks by Mexico-Based Transnational Criminal Organizations to Launder Illicit Proceeds.”
<https://www.fincen.gov/resources/advisories/fincen-advisory-fin-2025-a003>
2. FinCEN Advisory PDF (FIN-2025-A003). <https://www.fincen.gov/system/files/2025-08/FinCEN-Advisory-CMLN-508.pdf>
3. Chainalysis, “2026 Crypto Crime Report: Introduction” (Jan. 2026) (stablecoins share of illicit transaction volume). <https://www.chainalysis.com/blog/2026-crypto-crime-report-introduction/>
4. TRM Labs, “2025 Crypto Crime Report” (Feb. 10, 2025) (2024 chain distribution; TRON share). <https://www.trmlabs.com/reports-and-whitepapers/2025-crypto-crime-report>
5. Public Law 119–27 (GENIUS Act), July 18, 2025 (PDF).
<https://www.congress.gov/119/plaws/publ27/PLAW-119publ27.pdf>
6. U.S. Department of the Treasury, “GENIUS Act Implementation” (Federal Register notice, Sept. 19, 2025).
<https://www.federalregister.gov/documents/2025/09/19/2025-18226/genius-act-implementation>



7. SSA petition announcement page (Dec. 8, 2025). <https://fedmsb.org/ssa-files-multi-agency-petition-urging-coordinated-genius-act-implementation-heightened-usdt-risk-treatment-and-on-chain-node-kyc-rules/>
8. PR Newswire syndication of SSA petition announcement (Dec. 2025). <https://www.prnewswire.com/news-releases/ssa-petitions-us-regulators-to-align-genius-act-implementation-issue-usdt-high-risk-guidance-and-establish-on-chain-node-kyc-302635404.html>
9. FATF–Egmont Group, “Trade-Based Money Laundering: Trends and Developments” (2020) (PDF). <https://www.fatf-gafi.org/content/dam/fatf-gafi/reports/Trade-Based-Money-Laundering-Trends-and-Developments.pdf>
10. FATF, “Trade-Based Money Laundering Risk Indicators” (2021) (PDF). <https://www.fatf-gafi.org/content/dam/fatf-gafi/reports/Trade-Based-Money-Laundering-Risk-Indicators.pdfqa>